

Yirong Qiang

University of Electronic Science and Technology of China | Email: yirongqiang1@gmail.com

Education Background

September 2022- June 2026 University of Electronic Science and Technology of China (Project 985) BEng

Yingcai Honors College (the only honors college of UESTC for selected talented students)

Major: Artificial Intelligence & Mathematics (Yingcai Honors Program of UESTC)

GPA :4.00/4.00 Rank: 1 Weighted Average Score: 92.46

English Proficiency: IELTS 7.0, CET-4: 603, CET-6:522, proficient academic English skills

Key courses and Grades:

Advanced Calculus III: 98 | Comprehensive Experiments of AI: 97 | Multivariate Statistical Analysis: 97

Programming Design Methods (C/C++) : 93 | Principles of Computer Organization: 92 | Algorithms for Optimization: 94

Mathematics & MATLAB: 96 | English for Academic Reading: 95 | Theoretical Mechanics : 93

Publications and Preprints

Yirong Qiang, Lianlei Shan†. "DIWA: Decision-Influential World Abstraction for VLA-WAM Policies. " Submitted to **ICLR 2027**, under review.

Yirong Qiang, Jiahe Zhang, Ming Zhou, Yuxiu Pei, and Lianlei Shan†. "Stage-Verifiable Latent Chain-of-Thought with Local Repair for Long-Horizon Robotic Manipulation. " *International Conference on Machine Learning (ICML)*, 2026. **Accepted**.

Yirong Qiang*, Xi Hong*, Zhewei Li, Yuling Zheng, Yi Lu, and Lun Chen†. "Taming Multi-Agent Collaboration for 3D Situated Reasoning via Uncertainty Alignment. "*ICLR 2025 Workshop on Foundation Models in the Wild*, 2025.

Yunmeng Kui*, **Yirong Qiang***, Haoran Wu, Tusheng Ren, Wu Cheng, and Kang Zhao†. "OmniIntents: Enhancing Intent Prediction and Agent Selection through Real-World Multimodal Inputs and LLM Integration. " *ACM CHI Conference on Human Factors in Computing Systems (CHI)* , 2025.

Research Experience

Research: "DIWA: Decision-Influential World Abstraction for VLA-WAM Policies" -- **RA@Tsinghua University**

Research areas: *Embodied AI · World Action Models · Decision-making*

June 2026- September 2026

- **Overview:** Developed **DIWA** (Decision-Influential World Abstraction), a framework for efficient vision-language-action policies that selects decision-relevant future queries before world-model decoding. Combined intervention-based influence estimation with candidate-action regret geometry to preserve useful predictive information while reducing inference computation.

Results: Improved task success across simulation benchmarks and real-robot manipulation, with stronger robustness to appearance changes, background motion, and contact counterfactuals. Substantially reduced inference latency through selective future decoding while maintaining performance comparable to dense imagination. Ablation studies supported the contributions of influence supervision and regret geometry.

Achievements: "DIWA: Decision-Influential World Abstraction for VLA-WAM Policies " was submitted to ICLR 2027.

- **Personal Contribution:** As **first author**, led the research direction and methodological design. Formalized the finite-candidate decision-equivalence property and led the analysis of benchmark results, budget tradeoffs, ablations, and mechanism diagnostics.

Research: "Stage-Verifiable Latent Chain-of-Thought with Local Repair for Long-Horizon Robotic Manipulation"

Research areas: *Embodied AI · VLA models · Latent Reasoning*

October 2025- January 2026

- **Overview:** Developed **VLCOT**, a VLA framework for long-horizon robotic manipulation that aligns latent reasoning with observable task stages. The framework verifies execution progress and selectively regenerates affected plan segments while retaining valid prior reasoning.

Results: Achieved strong performance on long-horizon manipulation benchmarks and improved recovery from execution failures. Local repair reduced inference time compared with full replanning, demonstrating an effective trade-off between computational efficiency and task recovery.

Achievements: "Stage-Verifiable Latent Chain-of-Thought with Local Repair for Long-Horizon Robotic Manipulation" was accepted by ICML 2026.

- **Personal Contribution:** As **first author**, contributed to problem formulation and led the core method design and model implementation, including stage alignment, observation-based verification, and dependency-aware local repair. Conducted most benchmark evaluations and led manuscript writing.

Research: "Taming Multi-agent Collaboration for 3D Situated Reasoning" -- **Intern@AIR, Tsinghua University**

Research areas: *Agentic AI(embodied agents) · Multimodal Reasoning · 3D vision(3D scene understanding)*

December 2024-March 2025

- **Overview:** The research aims to build a multi-agent collaboration framework for the 3D Situated Question Answering (SQA3D) task. The study integrates large multimodal models and Conformal Prediction calibration to design a reasoning system, explicitly models the semantic understanding and embodied

reasoning processes of 3D scenes, providing a feasible and interpretable path for embodied agents in real-world scenarios.

Results: The model outperformed state-of-the-art end-to-end models in the SQA3D benchmark, improving situational reasoning accuracy. The research alleviates the strong dependency on domain-specific data and improves generalization in zero-shot scenarios.

- **Achievements:** "Taming Multi-agent Collaboration for 3D Situated Reasoning via ..." was published by ICLR 2025 workshop (foundation models in the wild).
- **Personal Contribution:** As **co-first author**, I was responsible for most of the core framework pipeline construction, code, designing the multi-agent system architecture. I conducted the benchmark experiments and contributed to the sections on related work, methods, and experiments in the paper.

Research: "OmniIntents: Enhancing Intent Prediction and Agent Selection through Real-World Multimodal Inputs and LLM Integration" -UESTC

Research areas: *Human-Computer Interaction · Agentic AI · Multimodal AI*

June 2024- November 2024

- **Overview:** Involved in human-computer interaction (HCI) and multimodal large language model (MLLM) research, exploring the recognition and response to human intentions through in-context learning, chain-of-thought (CoT), and fine-tuning. The study used multimodal inputs from video datasets to enhance real-time interactive systems with human intent prediction and task planning, which was implemented in AR/VR environments.
Results: The developed system demonstrated excellent performance in handling complex multi-channel inputs (language, text, visual), significantly improving task planning accuracy and execution efficiency, addressing misunderstandings caused by inconsistent modal inputs.
- **Achievements:** "OmniIntents: Enhancing ...through Real-World Multimodal Inputs and LLM Integration," was published by CHI-25 conference.
- **Personal Contribution:** As **co-first author**, I conducted literature research on multimodal HCI, built the core framework pipeline, and performed training and experiments, with some support from a PhD student for the AR system design and data collection.

Project Experience

March 2024-June 2024 Trustworthy Agent Socioeconomic System Simulation Based on Large Language Models – Research Assistant

- **Overview:** Focused on AI & computational social science research, particularly **multi-agent** modeling and simulation of complex socioeconomic systems. The research innovatively used LLMs to simulate agent behaviors and the emergent processes in macro-level phenomena. The simulation helped observe the dynamics of policy effects.
- **Achievements:** The project won second place (First Prize) at the New Engineering Exhibition.
- **Personal Contribution:** Designed and implemented multi-layer agent architectures, and built an expandable framework based on LLMs.

Summer Camp(Excellent Camper/Early Admission)

July 2025 Excellent Camper of Beijing Institute for General Artificial Intelligence (BigAI) ,Pengcheng National Laboratory (PCNL)

July 2025 Excellent camper, School of Artificial Intelligence(SAI), Shanghai Jiao Tong University

August 2025 Selected for Early Admission, **Tsinghua** Shenzhen International Graduate School

Selected Honors & Awards

October 2023: **National Scholarship**

October 2023 & October 2024: **Academic Excellence Star**, UESTC

November 2023: Awardee of the Silver Award, **ACM-ICPC** Regional Invitational Contest, Chengdu Site

November 2020: **Provincial First Prize**, National High School Mathematics Olympiad; Provincial First Prize, National High School Physics Olympiad; Provincial Second Prize, National Chemistry Olympiad

Leadership/Work Experience

December 2023 – March 2026 **Co-founder**, UESTC AI Society

- Co-founded an AI community at the University of Electronic Science and Technology of China, now comprising **2,000+ members**.
- Helped connect community members with AI researchers and contributed to an open-source AI community.

June 2026 – September 2026 **RA, CST, Tsinghua University**

December 2024 – March 2025 **Intern, AIR, Tsinghua University**

September 2023 – September 2024 **Minister**, Elite Student Development Center, UESTC

September 2022 – September 2023 **Class Monitor**, Yingcai Honors Program of UESTC

Other Qualifications

AI: Proficient in Python, MATLAB, LaTeX, and PyTorch deep learning framework. Solid foundation in deep learning.

Personality Traits: Highly adaptable and quick to learn. Actively explores frontier research and presents proposals. Capable of handling multiple research projects, engineering competitions, and coursework simultaneously, with strong stress resilience and project management skills.